Assignment 2: Proposed Approach and Experimental Design

SemEval 2027 / ICASSP 2027 Shared Task / Challenge Project

Purpose: move from task and data understanding to a justified, reproducible experimental system.

Assignment 2 develops the proposed method, comparison systems, evaluation plan, experimental protocol, and initial implementation. It should build directly on the evidence and baselines established in Assignment 1.

Expected progression: Proposed method → Justification → Comparison systems → Evaluation design → Experimental protocol → Initial implementation → Preliminary verification.

1. Proposed Method

Present a clear and technically justified approach for solving the shared task. The description must be specific enough that another researcher could implement the system.

- Describe the system, its components, and the flow from input to prediction.
- State the input, output, prediction target, and expected model behavior.
- Explain the role of the features, representations, or modalities identified in Assignment 1.
- Specify the model architecture, training objective, loss function, and prediction strategy where applicable.
- Explain how the method addresses imbalance, domain or speaker variation, noisy data, long inputs, cross-lingual variation, or multimodal variation.
- Document preprocessing, augmentation, normalization, feature selection, and sampling decisions.
- State expected limitations and likely sources of error.

Required method diagram: include raw input, preprocessing, feature or representation extraction, model components, prediction layer, and output/evaluation. Identify trainable and frozen components.

2. Justification of the Proposed Approach

Explain why the proposed method is appropriate for this particular task and dataset. Connect Assignment 1 findings, dataset properties, prior work, and expected model behavior.

Evidence from Assignment 1	Design decision	Expected effect
Observed data or task pattern	Component, representation, or training choice	Expected performance or error-behavior consequence
Example: class imbalance	Use class-weighted loss and macro-F1	Improve minority-class sensitivity and reduce majority-class dominance

Use the following reasoning pattern for important decisions: observation → interpretation → modeling implication. Do not justify a method only by stating that a model is modern or widely used.

3. Comparison Systems

Define systems that allow the proposed method to be evaluated fairly and interpreted meaningfully. Use identical data splits and evaluation procedures wherever possible.

3.1 Simple or majority baseline

- Majority-class or empirical-distribution prediction for classification.
- Mean or median prediction for regression.
- A simple retrieval or heuristic system where appropriate.
- Explain what the baseline reveals about task difficulty and imbalance.

3.2 Classical machine-learning baseline

- Use interpretable, task-relevant features from Assignment 1.
- Possible models include Logistic Regression, Linear SVM, Naive Bayes, Random Forest, Gradient Boosting, XGBoost, or LightGBM.
- Report preprocessing, feature representation, hyperparameters, feature selection, and class weighting or resampling.

3.3 Pretrained-model baseline

- Select a suitable pretrained encoder, such as BERT-family or XLM-R for text, or wav2vec 2.0, HuBERT, WavLM, or Whisper representations for speech.
- State whether representations are frozen, partially fine-tuned, fully fine-tuned, or used through prompting.
- Document the prediction head and training procedure.

3.4 Proposed system

Explain which additional component, hypothesis, representation, or training strategy distinguishes the proposed system from the comparison systems.

4. Ablation Plan

Design experiments that isolate the contribution of important components. Each ablation must answer a meaningful research question.

Ablation	Hypothesis	Expected interpretation
Remove a feature group or modality	The component contributes task-relevant information	Performance change estimates its contribution
Frozen encoder versus fine-tuned encoder	Task adaptation improves generic representations	Compare task fit against computational cost
With versus without class weighting or augmentation	Training adjustments improve robustness	Inspect minority or difficult-group metrics

Possible ablations include metadata removal, alternative pooling strategies, text-only versus speech-only versus multimodal input, and different preprocessing or normalization methods.

5. Evaluation Measures

5.1 Official metric

State the official shared-task metric and explain its mathematical definition, what it rewards or penalizes, whether higher or lower is better, and how it behaves under imbalance, skew, or unequal group sizes. Identify macro, micro, weighted, or sample averaging where applicable.

5.2 Secondary metrics

Include complementary measures such as accuracy, precision, recall, macro-F1, weighted-F1, balanced accuracy, ROC-AUC, PR-AUC, MAE, RMSE, Pearson or Spearman correlation, WER/CER, BLEU, ROUGE, BERTScore, or task-specific metrics as appropriate.

5.3 Subgroup and robustness analysis

- Report performance across labels, languages, speakers, domains, sources, recording conditions, document lengths, and relevant metadata groups.
- Discuss whether aggregate performance hides systematic weaknesses in particular subsets.

6. Experimental Protocol

Specify the complete protocol that will be used for all experiments.

- Dataset version and source; official train, development, and test partitions; and any additional validation split.
- Preprocessing, tokenization or segmentation, feature extraction, and augmentation.
- Model initialization, epochs or stopping criteria, batch size, optimizer, learning rate, scheduling, regularization, and dropout.
- Random seeds, hardware, software environment, repeated runs, model-selection criterion, and hyperparameter-search strategy.
- Checkpoint selection, handling of missing or duplicated instances, class-imbalance handling, and explicit prevention of data leakage.
- Distinguish local development evaluation from official hidden-test or server evaluation.

7. Statistical Reliability and Reproducibility

- Run important experiments with multiple random seeds where feasible.
- Report mean and standard deviation, and confidence intervals or significance tests where appropriate.
- Use identical data splits across comparison systems.
- Keep preprocessing and evaluation scripts fixed across experiments.
- Record configurations, save checkpoints and predictions, and preserve the exact code used for reported results.

Do not claim superiority based on a very small numerical difference without considering variance and experimental uncertainty.

8. Initial Implementation

Implement a minimal but complete version of the proposed system. It need not be optimized, but it must run end-to-end on a manageable subset or the complete development data.

- Data loading and preprocessing.
- Feature or representation extraction.

- Model definition and training or inference procedure.
- Evaluation script and required prediction format.
- Configuration file or clearly documented parameters.
- Checkpoint saving and loading.
- Reproducible random seeds.

Minimum implementation requirement: demonstrate that data load correctly, inputs are transformed into the intended representation, the model can train or run inference, predictions can be generated, at least one metric can be computed, and the experiment can be repeated from the documented configuration.

9. Preliminary Verification

Run a small initial experiment to verify that the proposed system and evaluation pipeline are operational. Report the data split and number of examples, model configuration, training duration or computational cost, initial metrics, warnings or failures, evidence of meaningful predictions, and issues to resolve before the full experiment.

Preliminary results verify implementation and pipeline correctness; they should not be presented as final performance claims.

10. Required Report Structure

1. Proposed Approach
2. Method Justification
3. System Architecture
4. Comparison Systems
5. Ablation Plan
6. Evaluation Measures
7. Experimental Protocol
8. Initial Implementation
9. Preliminary Results
10. Reproducibility and Limitations
11. Planned Experiments

Use LaTeX/Overleaf with clear sectioning, mathematical notation where necessary, tables for system comparisons and configurations, figures for the proposed architecture and workflow, BibTeX references, and cross-references for tables and figures.

Assignment 2 Submission Checklist

☐ Clearly specified proposed method

☐ Justification grounded in Assignment 1 findings and relevant literature

☐ Proposed-system architecture diagram

☐ Majority or simple baseline

☐ Classical machine-learning comparison system

☐ Pretrained-model comparison system

☐ Defined ablation experiments

☐ Official and secondary evaluation measures

☐ Complete experimental protocol

☐ Leakage prevention and data-split description

☐ Initial end-to-end implementation

☐ Preliminary verification experiment

☐ Reproducibility details, including seeds and configurations

☐ LaTeX/Overleaf report with tables, figures, and references

☐ Explicit limitations and planned next experiments

Core principle: naming a model is not a proposed approach. The assignment must demonstrate a defensible connection between task evidence, modeling decisions, comparison systems, evaluation measures, and a reproducible experimental protocol.